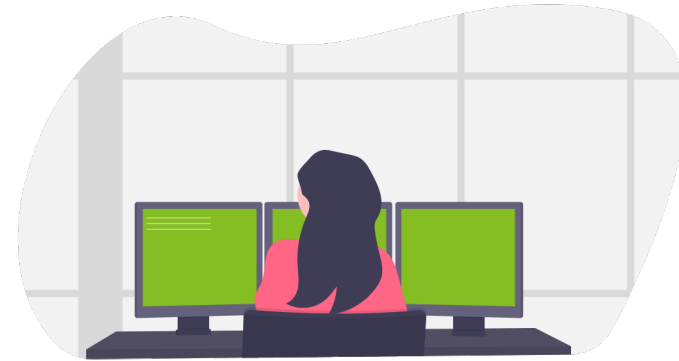
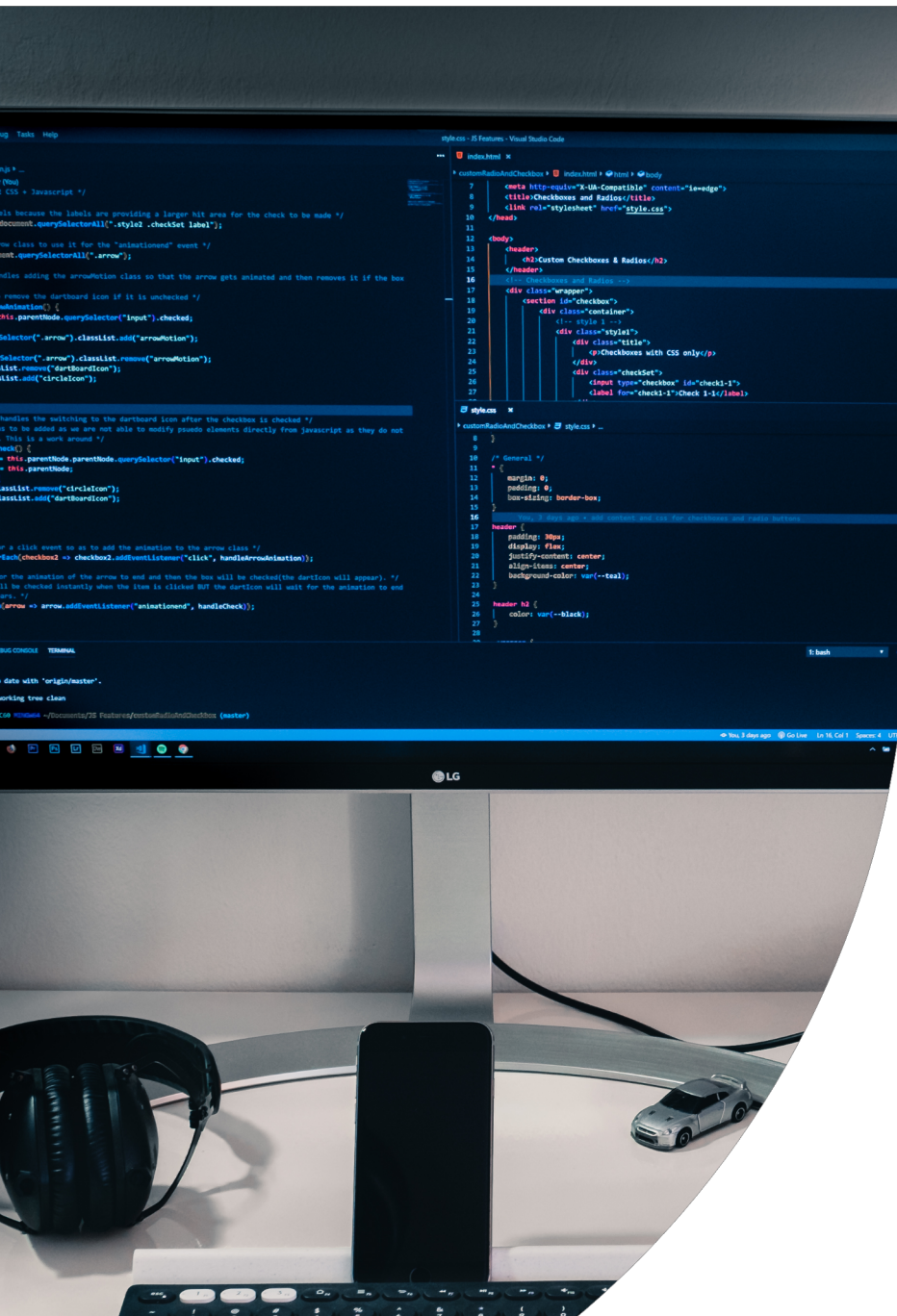


Software Development **Proposal**

Software Development Proposal



Software Development Proposal

Deloitte.

1. Overview

- *This project aims to develop a **Real-Time Manufacturing Status Dashboard**.*
- *The system will monitor machines across multiple factories and display their status in real time.*
- *It will help users track machine performance, detect issues instantly, and reduce downtime.*
- *The dashboard will be accessible only to authorised employees through a secure login system.*
- *The solution will improve operational efficiency and enable faster decision-making.*

2. Scope

The scope of this project involves the development of a real-time manufacturing dashboard that provides a centralized and structured view of machine health across multiple factories.

As illustrated in the wireframe, the system displays a hierarchical layout where factories are listed at a high level, and each factory can be expanded to reveal individual machines along with their operational status.

The dashboard enables users to monitor both factory-level and machine-level performance in real time.

Each factory is represented with a status indicator reflecting its overall condition, while individual machines display their health status (healthy or unhealthy) along with the latest update timestamp.

This design allows users to quickly identify problematic areas and navigate efficiently to detailed machine-level insights.

The system will support continuous data updates to ensure that all information displayed is current and reliable.

Users will be able to interact with the interface by expanding or collapsing factory views, enabling a clear and organized representation of complex manufacturing data.

In addition to real-time monitoring, the system will incorporate analytical capabilities to enhance operational decision-making.

Historical data will be stored and analysed to identify patterns such as frequent machine failures, performance degradation, and downtime trends. These insights will help in predictive maintenance and improving overall efficiency.

✓  Daikibo Factory Meiyo	Last update: <1min ago	◀
✓  Daikibo Factory Seiko	Last update: <1min ago	◀
✓  Daikibo Berlin	Last update: <1min ago	◀
✗  Daikibo Shenzhen	Last update: <1min ago	▼
✗  CNC	Last update: 2min ago	▼
✗  Status: Unhealthy	2min ago	
✓  Status: Healthy	12min ago	
Load More		
✓  LaserCutter	Last update: <1min ago	◀
✓  HeavyDutyDrill	Last update: <1min ago	◀
✓  SpotWelder	Last update: <1min ago	◀
✓  LaserWelder	Last update: <1min ago	◀
✓  MetalPress	Last update: <1min ago	◀
✓  Furnace	Last update: <1min ago	◀
✓  ConveyorBelt	Last update: <1min ago	◀
✓  AirWrench	Last update: <1min ago	◀

The system will include the following features:

- *Real-time monitoring of machine status across all factories*
- *Hierarchical dashboard view (factory → machines)*
- *Status indicators with clear visual representation (healthy/unhealthy)*
- *Timestamp display for latest machine updates*
- *Expandable and interactive interface for easy navigation*
- *Historical data storage for tracking performance over time*
- *Basic analytics for identifying trends and recurring issues*
- *Alert mechanism for detecting abnormal machine behaviour*
- *Secure access through internal authentication system*

The system will also include a real-time alert mechanism that notifies users whenever a machine experiences abnormal behaviour or failure.

This ensures quick response and minimises downtime. Overall, the solution is designed to provide improved visibility, efficient monitoring, and data-driven decision-making within the manufacturing environment.

Software Development **Proposal**

3. Estimate

*The estimated effort required to complete this project is approximately **120–140 man-hours**. This estimate includes the time required for development, testing, and integration of the system within the client's intranet environment. The development phase will require the majority of the effort, as it involves building the frontend dashboard, backend services, real-time data handling, and user authentication system. Testing will ensure that the system functions correctly under different scenarios and that real-time updates and alerts work reliably. Integration will involve deploying the system within the client's internal network and ensuring compatibility with existing infrastructure.*

Effort Breakdown:

- **Development:** 80–90 man-hours
(Frontend dashboard, backend logic, database setup, real-time data processing, authentication)
- **Testing:** 25–30 man-hours
(Functional testing, performance testing, bug fixing)
- **Integration:** 15–20 man-hours
(Deployment on intranet, system configuration, final validation)

4. Timeline

5th May 2026 – Design phase begins

8th May 2026 – System architecture and UI design finalized

9th May 2026 – Development phase begins

18th May 2026 – Backend development completed

22nd May 2026 – Frontend dashboard development completed

23rd May 2026 – System integration begins

25th May 2026 – Testing phase begins

28th May 2026 – Testing and bug fixing completed

30th May 2026 – Deployment on client intranet

1st June 2026 – Final review and project completion

5. Support

The system will be supported through continuous maintenance and monitoring to ensure its reliable and efficient operation after deployment.

The development team will provide timely resolution of any bugs or technical issues that may arise, minimizing disruption to users.

Regular updates and improvements will be implemented to enhance system performance and incorporate user feedback.

The system will also be monitored to ensure that real-time data processing and alert mechanisms are functioning accurately.

In addition, the solution is designed to be scalable, allowing for the integration of additional factories and machines as required in the future.

This ensures that the system remains adaptable to evolving business needs. Overall, continuous support will be provided to maintain system stability, improve functionality, and ensure long-term usability.